HARDWARE WORKSHOP

Q(8) : WHAT IS A BOOTABLE DEVICE ?

WHY DO WE USE IT ?

COMMAN SOFTWARES TO CONVERT A DEVICE TO BOOTABLE DEVICE.

TALK ABOUT PERIPHERAL DEVICE.

Presented by ~ GOPI VAMJA

PRIYANSHI PATEL

SMIT VARASADA

SOHAM SINH.

What is a bootable device ?

- A bootable device is any storage device that contains the files needed to start a computer's operating system.
- When you power on a Computer , it looks for a bootable device to load the OS (like windows or Linux) into memory.
- Bootable devices contains the essential files and instructions such as OS and drivers which we need to start up a computer or any device.

Examples of bootable device : Hard disk / SSD

Pen drive

CD/DVD

What makes a device bootable ?

- A device is bootable if :
- It contains a bootloader (a small program that starts the operating system).
- It has boot sector stored in a specific area of the device.
- The device is properly formatted with a supported file system.
- The computer's BIOS/UEFI is set to detect and allow booting from that device.

Why we use it ?

- We use bootable devices because they allow a computer to start an operating system (OS) or utility software when the internal hard drive cannot or should not be used.

- MAIN REASONS FOR USING BOOTABLE DEVICES :
- To install an operating system.
- To start a computer when the main OS fails.
- If the system doesn't start due to a virus attack , crash or any other reason , a bootable device helps start the computer and fix the issue.

- Used to repair startup errors, such as : Recover deleted or lost data , restore system backups.
- It is also used to remove viruses and check security through scanning tools.
- To test or run an OS without installing it (Live OS).
- Hardware bootable tools help test : RAM , CPU, Hard Disk and other hardware components.
- Bootable devices are used to format or manage disks safely when the OS is not running.

Common software to convert a device to bootable device :-

- Here are common and widely used software tools to convert a device (USB/DVD) into bootable devices :

Rufus

BalenaEtcher

Ventoy

UNetbootin

YUMI

PowerISO

RUFUS :

~The primary function of Rufus is to create bootable USB drives, allowing you to install operating systems or run live environment on computers.

~It simplifies the process of transferring ISO images for operating systems like Windows, Linux, or recovery tools onto USB devices.

~ It is most popular for windows.

~It is fast , simple and free to use.

~It supports windows , Linux , ISO files

Unetbootin:

~UNetbootin (Universal Netboot Installer) is a cross-platform utility that can create live USB systems and can load a variety of system utilities or install various Linux distributions and other operating systems without a CD.

~ It works on windows , Linux , mac OS.

~It is mainly used for Linux distributions.

~It is a also free to use.

Balena Etcher:

~It is best for simple flashing of OS images like Linux to USBs or SD cards.

~It runs natively on Windows , macOS, and Linux.

Ventoy :

~Ventoy is an open-source tool used to create bootable USB drives for ISO , WIM,IMG,VHD and EFI files without needing to reformat the drive for every new image.

~It allows users to simply copy files onto the USB and select them from a boot menu , supporting multiple operating systems like windows , Linux ,Unix and both legacy BIOS and UEFI.

Peripheral Devices :-

- A peripheral device is any external device connected to a computer that adds functionality but is not part of the core computer system.
- These devices typically handle input , output (I/O) devices.
- While not essential for a computer to perform its basic operations , peripherals enhance the user's experience by expanding the system's capabilities.
- Common examples include keyboards < mice , printers and external drives.
- Although the term is sometimes used loosely,it generally refers to devices located outside the computer case.

PERIPHERAL DEVICES

Input Devices :

~An input device is defined as a device that converts incoming data and instructions into a pattern of electrical signals in binary code that are comprehensible to a digital computer.

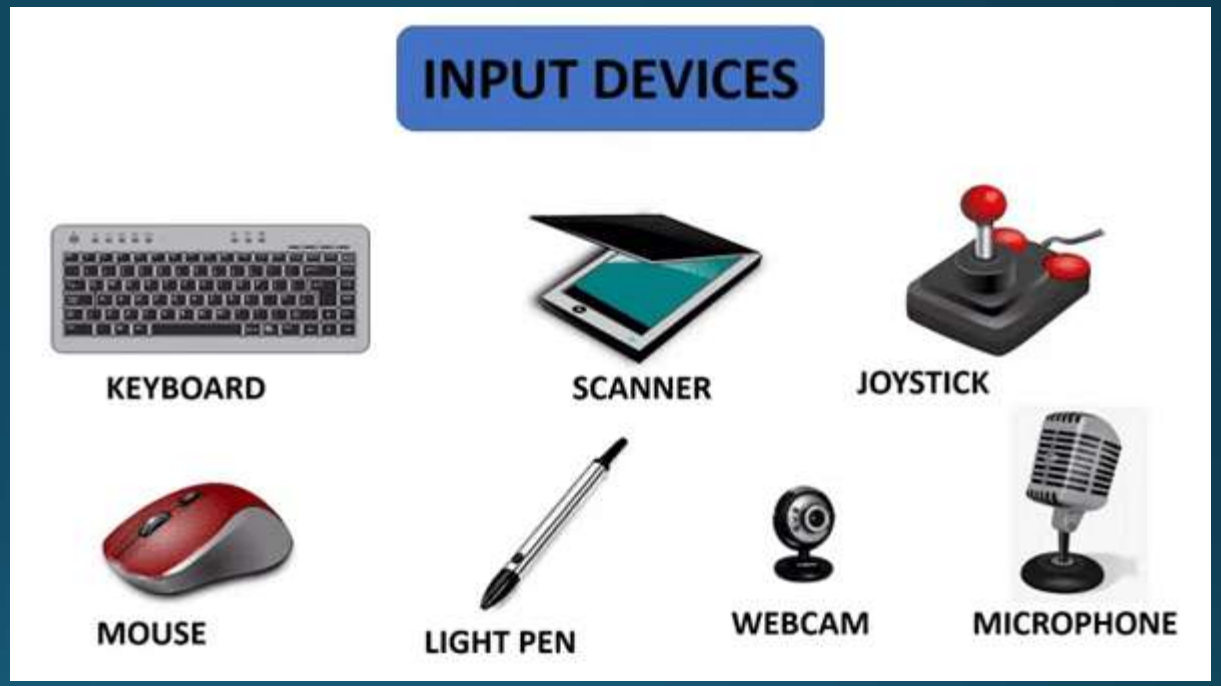
Like :-

Keyboard – enter text and command.

Mouse – control the cursor.

Scanner – convert physical documents and images into digital files

Microphone – an input device that allows users to record audio.



Output Devices :

~ An output device is generally the reverse of the input process and generally translates the digitized signals into form intelligible to the user.

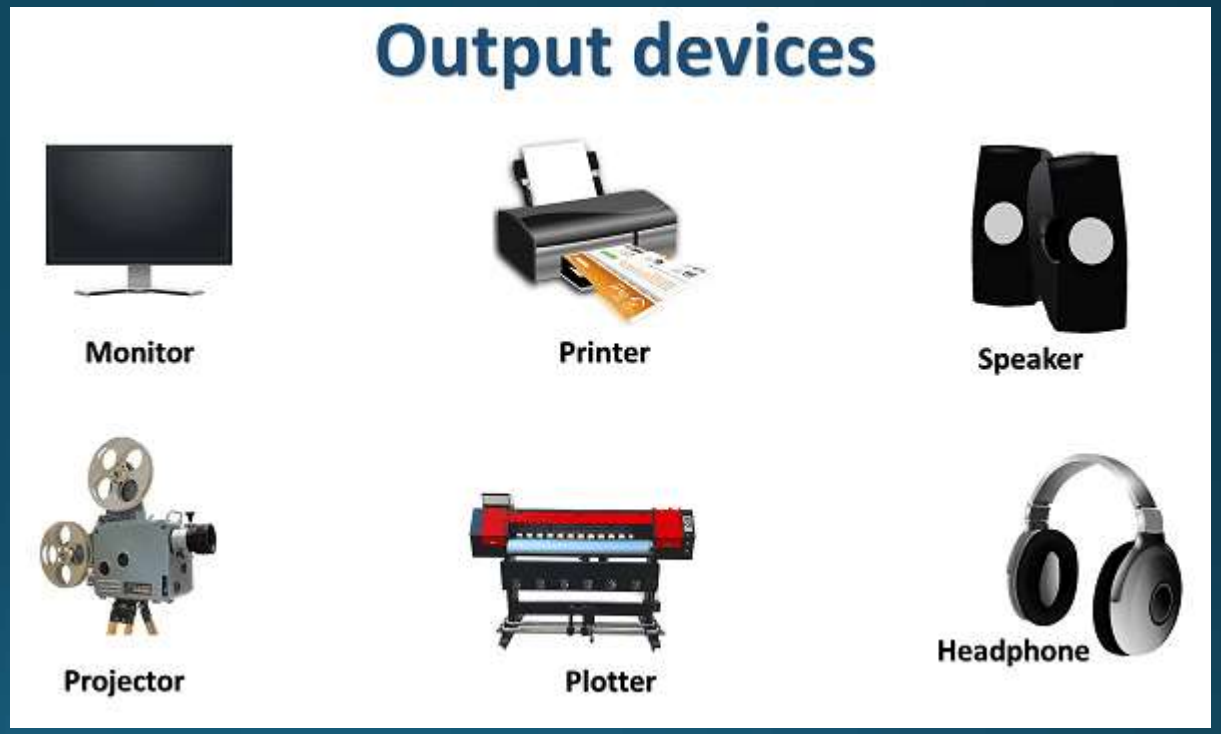
~The output device is also performed for sending data from one computer system to another.

Like:-

Monitor – displays the visual information.

Printer – produces physical copies of documents or images.

Speaker– produce audio.



Storage Device :

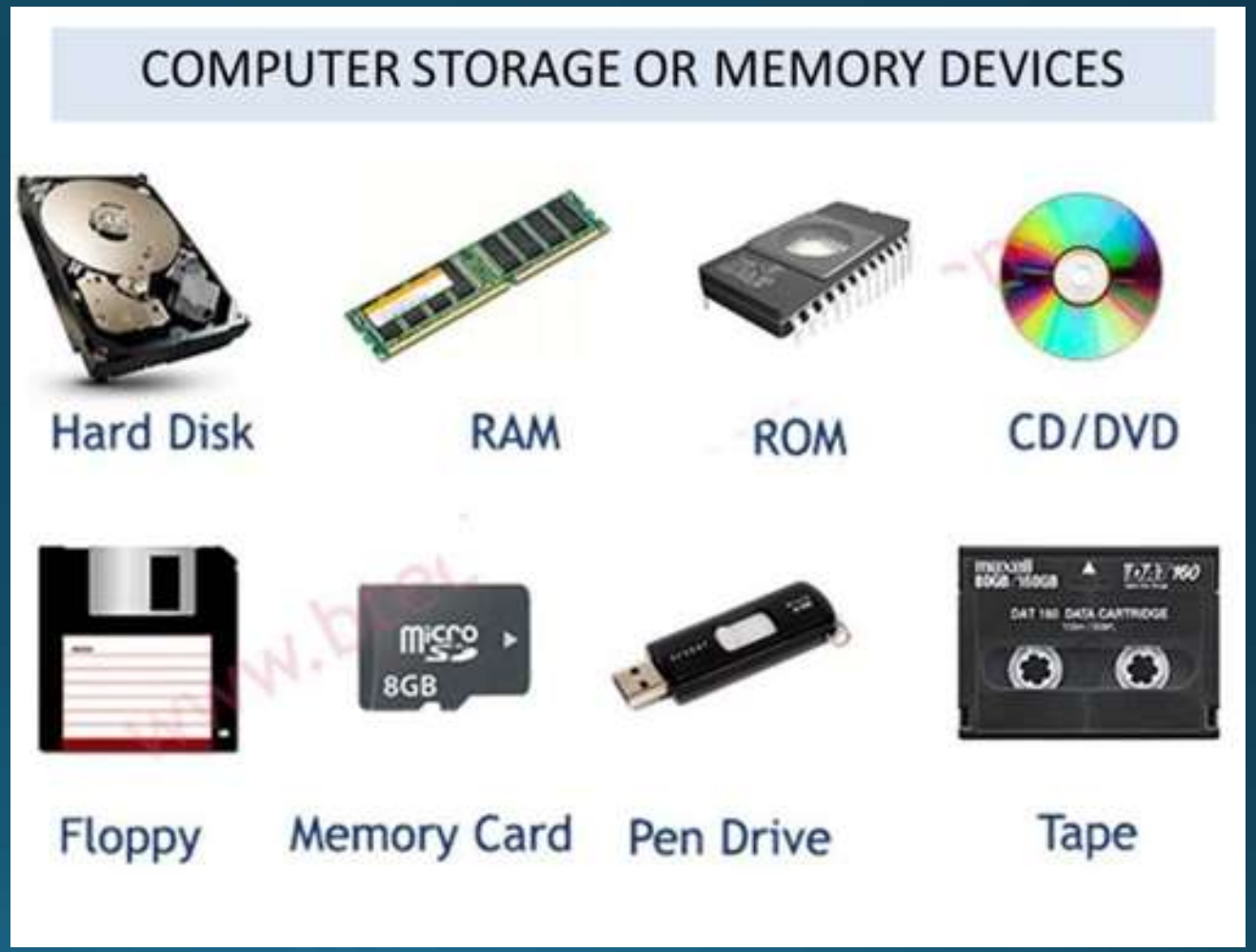
~ Storage device are used to store data in the system which is required for performing any operation in the system .

Like:-

Hard Drive – stores data and files on a computer system.

USB Drive – portable storage device that connects to a computer system to provide additional storage space.

Memory Card– that is commonly used in digital cameras and smartphones.



Communication Devices :-

- Communication devices are hardware devices that enables data exchange between computer systems or networks.

~Modem – allows a computer system to connect to the internet.

~Network Card – allows a computer system to connect to a network.

~Router – multiple devices to connect to a network.

Importance of peripheral devices:-

- Peripheral devices are important for enhancing functionality of a computer.
- Here is why peripheral devices are so importance.

1) Enhanced Functionality : Peripheral Devices expand a computer's capabilities, enabling it to perform various tasks.
2) User Interaction : Input devices like keyboards and mice allow users to control and interact with the system.
3) Data Output : Output devices such as monitors and printers displays or present processed data.

THANK YOU